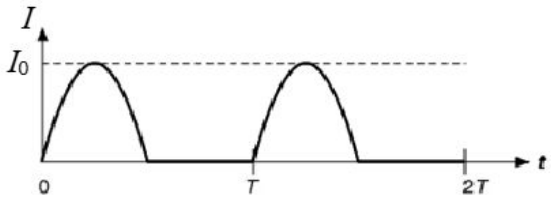
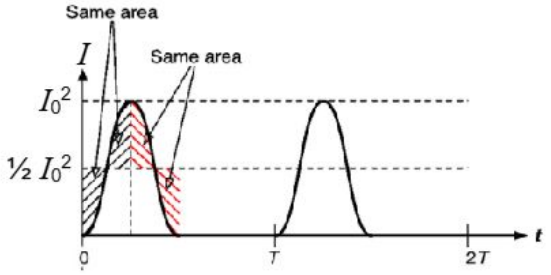
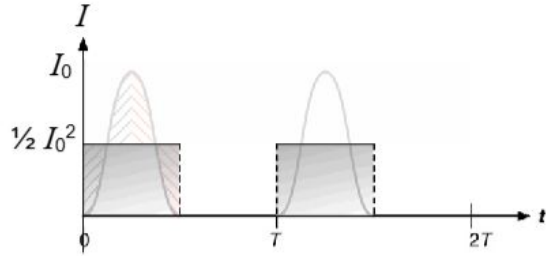
Since the separation of the plate is greater at the top, the electric field strength will be weaker and lines more separated.

21	**A**	Recall that to find the root-mean-square (r.m.s.) value of any waveform, **square** the values of the waveform, find their **mean**, and then take the **square root** of the mean. For half wave sinusoidal wave,  Step 1: square the values  Step 2: mean = ¼ I_0^2 Note that the ½ I_0^2 labelled on the graph is only the average value for half a period. The average value across the entire period is therefore ¼ I_0^2.  Step 3: square root of the mean = ½ I_0 Hence, **for half wave sinusoidal wave, rms current = ½ peak current** **For square wave, rms current = peak current** If the same power is dissipated, that means their rms currents are equal. Hence, $\frac{\text{peak current of square wave}}{\text{peak current of half wave rectified sinusoidal wave}} = \frac{1}{2}$
22	**A**	Assume A has a high potential and B has low potential. Let's label V_1, V_2, V_3, etc for points of potential where V_1 is the highest